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MA3521 — Variation of Parameters

Name:

Solve $\mathbf{y}' = A\mathbf{y} + \mathbf{f}(t)$ $\quad A = \begin{pmatrix} 1 & -3 \\ 3 & 1 \end{pmatrix}$ $\quad \mathbf{f}(t) = \begin{pmatrix} e^t \sec(3t) \\ 0 \end{pmatrix}$

1. $\mathbf{y}_c =$ $\qquad \Phi(t) =$

2. $\det \Phi(t) =$ $\qquad \Phi(t)^{-1} =$

3. $\mathbf{u}'(t) =$

Solve $\mathbf{y}' = A\mathbf{y} + \mathbf{f}(t)$ $\quad A = \begin{pmatrix} -5 & 3 \\ -8 & 5 \end{pmatrix}$ $\quad \mathbf{f}(t) = \dfrac{e^t}{1+t^2}\begin{pmatrix} 1 \\ 2 \end{pmatrix}$

1. $\mathbf{y}_c =$ $\qquad$ $\Phi(t) =$

2. $\det \Phi(t) =$ $\qquad$ $\Phi(t)^{-1} =$

3. $\mathbf{u}'(t) =$